



The Drift-Diffusion Model

BAMB! 2026
Module 1

Sequential Sampling Analysis

Decisions are *not* instantaneous but unfold in time



The Washington Post
Democracy Dies in Darkness

VATICAN FINDS GALILEO 'NOT GUILTY'

POPE ADMITS ERROR IN REJECTING THEORY

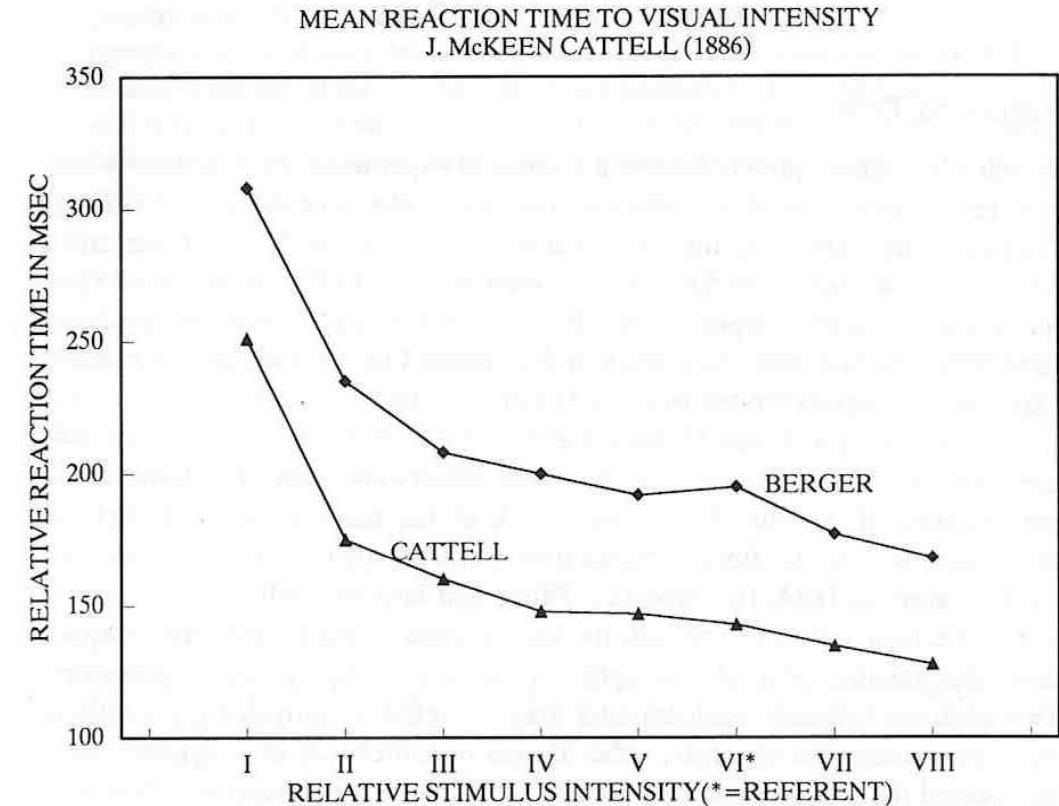
October 31, 1992

Why do we care?



Sequential Sampling Analysis

- Decisions unfold in time: information often comes sequentially (**sensory stream**), and actions are shaped by **timing constraints**
- The timing of decisions is a very valuable 'diagnostic' observable: *when* they are made can inform about *how* they are made.
- Some tasks show no variability in response accuracy (i.e. no errors)



Summary

1. A normative approach to sequential sample analysis
2. The Drift-Diffusion Model
3. Extensions, alternatives and limitations
4. Modelling behavioral data with the DDM



A principled way to sequential sample analysis

- Decide between two stimulus categories A or B based on noisy samples $x_1, x_2, x_3, \dots$
- We need:
 - a method to **accumulate evidence** regarding the current stimulus category over sensory samples
 - a **policy** to decide when to commit to a decision based on the accumulated evidence (and elapsed time)



$$\text{Posterior probability } p(A|x_1, x_2, \dots, x_N) = \frac{\overset{\text{Prior}}{p(A)} \overset{\text{Likelihood}}{p(x_1, x_2, \dots, x_N | A)}}{p(x_1, x_2, \dots, x_N)}$$

$$p(B|x_1, x_2, \dots, x_N) = \frac{p(B)p(x_1, x_2, \dots, x_N | B)}{p(x_1, x_2, \dots, x_N)}$$

Probability Ratio:

$$\frac{p(A|x_1, x_2, \dots, x_N)}{p(B|x_1, x_2, \dots, x_N)} = \frac{p(A)}{p(B)} \frac{p(x_1, x_2, \dots, x_N | H_1)}{p(x_1, x_2, \dots, x_N | H_1)}$$



Normative accumulation of evidence:

Samples x_i are independent

$$LLR_N = \log \frac{p(A|x_{1..N})}{p(B|x_{1..N})} = \log \left(\frac{p(A)p(x_1, x_2, \dots, x_N | A)}{p(B)p(x_1, x_2, \dots, x_N | B)} \right) =$$

$$\log \left(\frac{p(A)}{p(B)} \prod_{i=1}^N \frac{p(x_i | A)}{p(x_i | B)} \right) = \log \frac{p(A)}{p(B)} + \sum_{i=1}^N \log \frac{p(x_i | A)}{p(x_i | B)}$$

Accumulation
of evidence

Log Prior ratio

*Log likelihood ratio
= evidence e_i*

$$LLR_n = LLR_{n-1} + e_n$$

$$LLR_0 = \log \frac{p(A)}{p(B)}$$



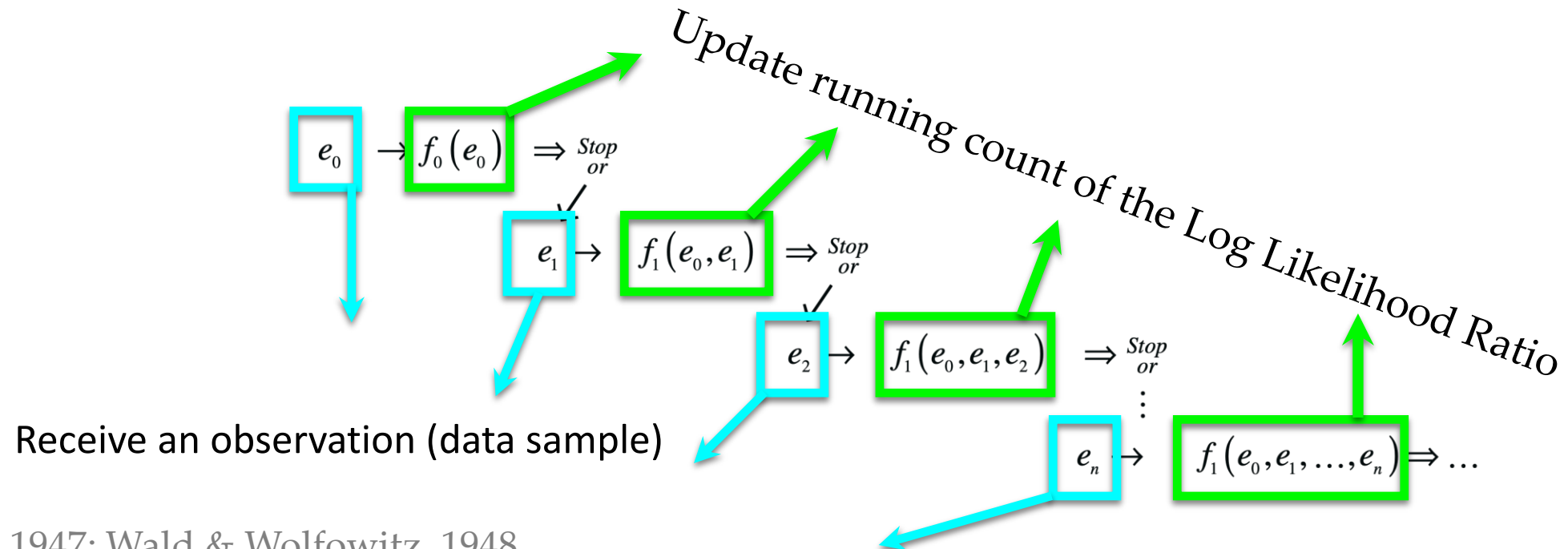
Sequential Probability Ratio Test (SPRT)

Evidence accumulation

$$\begin{aligned} LLR_n &= \log \frac{p(A)}{p(B)} + \sum_{i=1}^N \log \frac{p(x_i|A)}{p(x_i|B)} \\ &= e_0 + e_1 + e_2 + \dots e_n \end{aligned}$$

Decision policy

$$f(S_n) = \begin{cases} A & \text{if } LLR_n \geq a \\ B & \text{if } LLR_n \leq -a \\ \text{Wait} & \text{if } -a < LLR_n < a \end{cases}$$



From the SPRT to the DDM

Discrete time

noisy observations

$$\Delta LLR_n = \hat{e}_n = e_n + \sigma \eta_n$$

LLR update *evidence* *gaussian noise*

...until $|LLR_n|$ reaches a

Continuous time

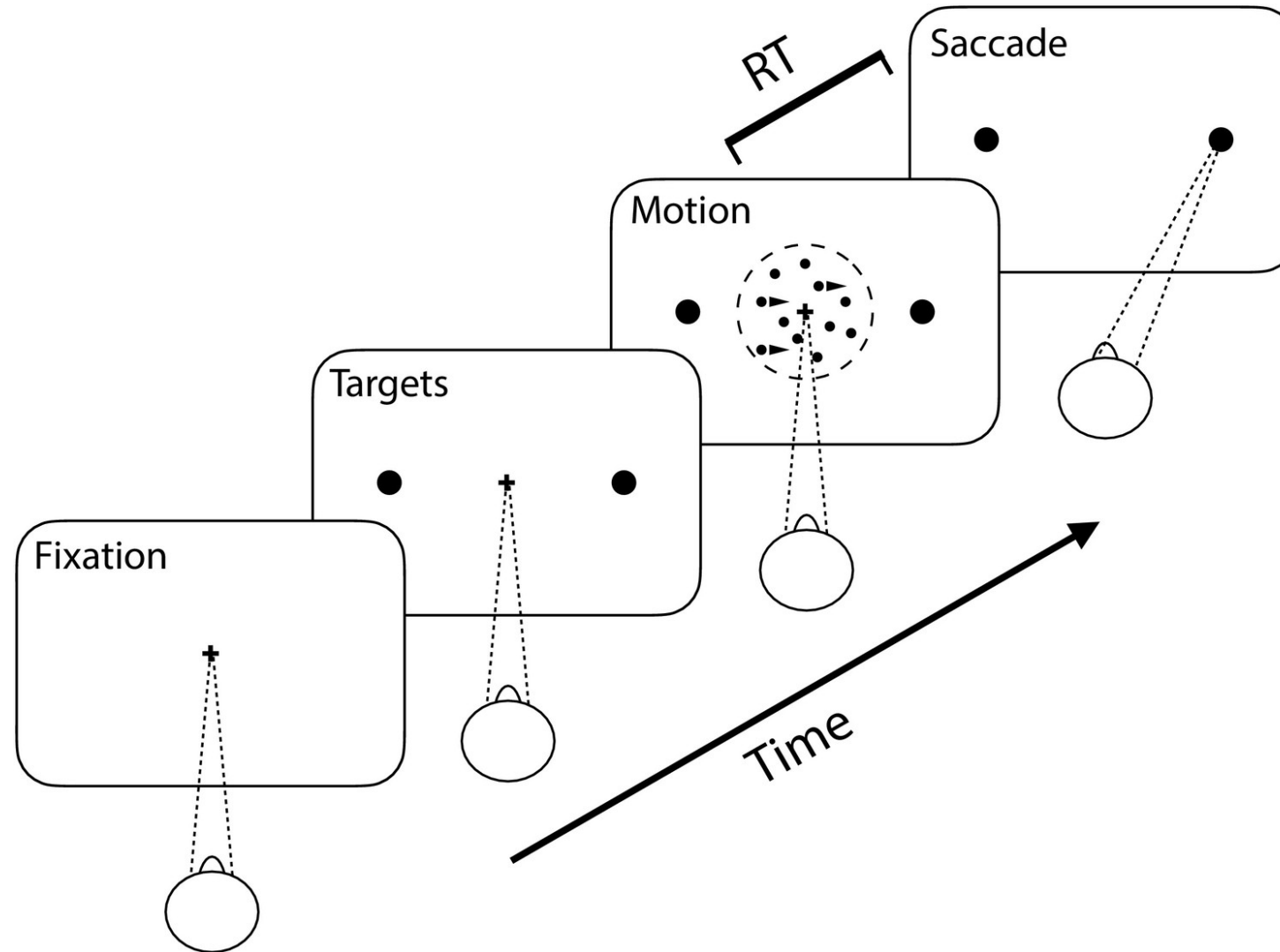

$$dx_t = \mu dt + \sigma dW_t$$

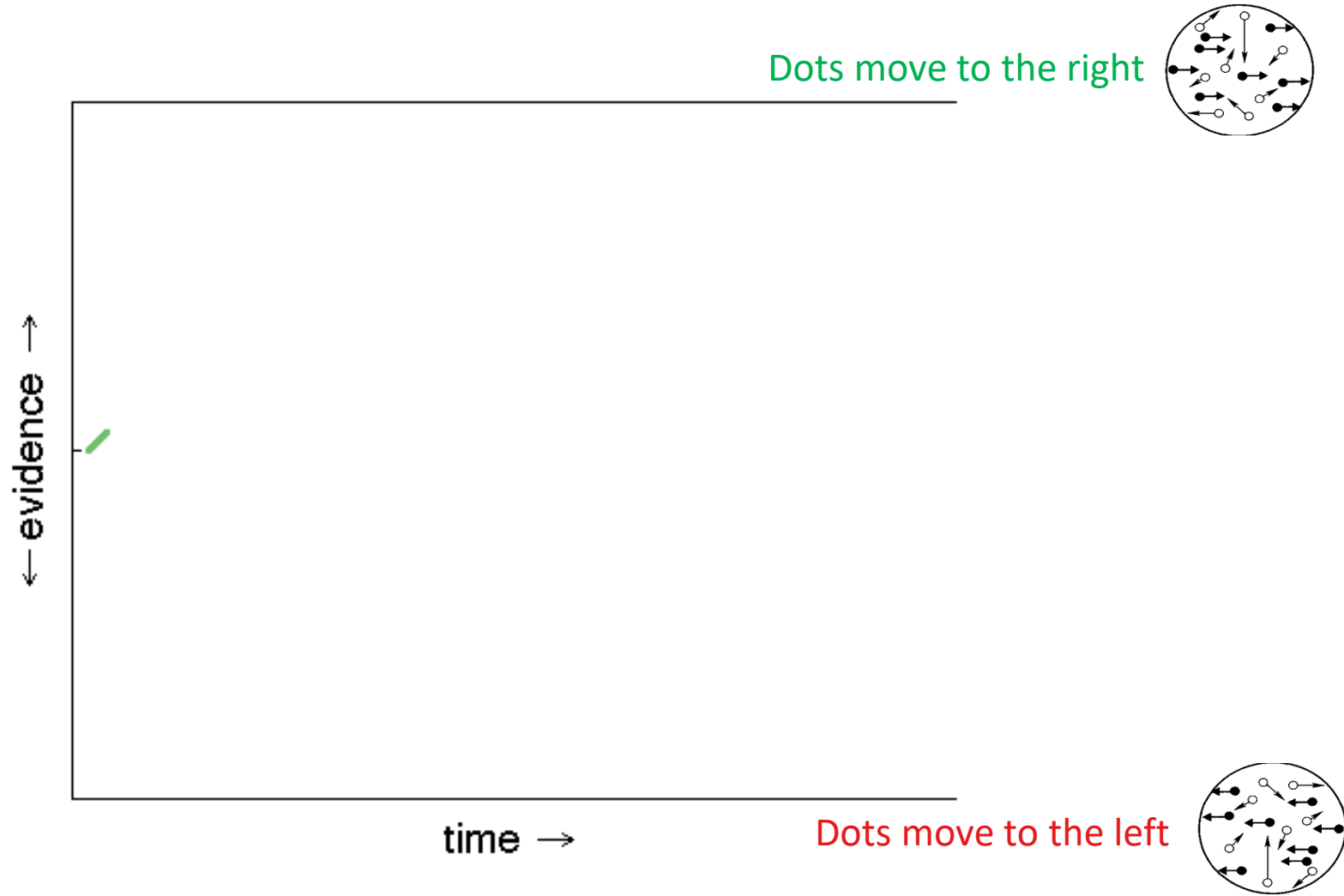
decision variable *drift (rate μ)* *Diffusion (Gaussian noise of variance $\sigma^2 dt$)*

...until $|x_t|$ reaches a

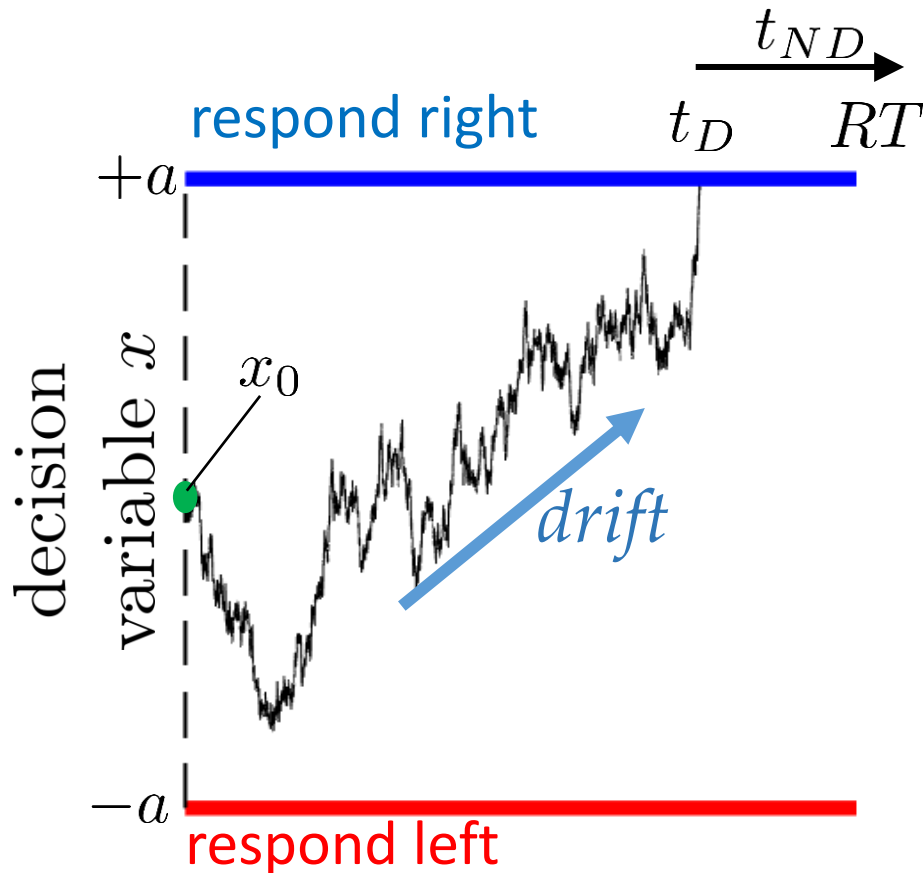


the random-dot-motion task





The Drift Diffusion Model



$$dx_t = \underbrace{\mu dt}_{\text{drift (rate } \mu)} + \underbrace{\sigma dW_t}_{\text{Diffusion (Gaussian noise of variance } \sigma^2 dt)}$$

dx_t is the decision variable.

initial conditions

$$x_0 = (2z - 1)a$$

termination

$$|x_{t_D}| = a$$

reaction time

$$RT = t_D + t_{ND}$$

PARAMETERS

- μ : drift rate
- σ : noise infinitesimal standard deviation
- a : threshold
- z : starting point (from 0 to 1)
- t_{ND} : non-decision time

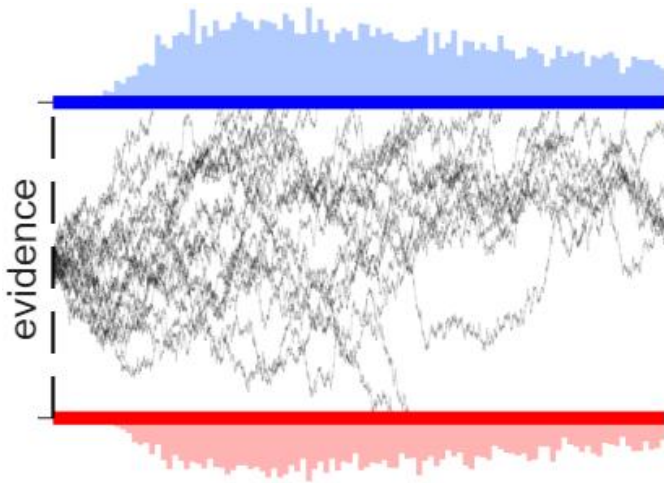


Inspect the impact of DDM parameters on behavior

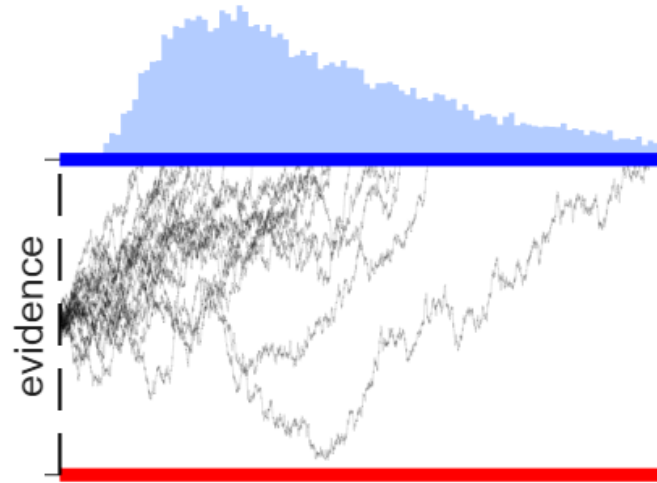


DDM parameters: the drift

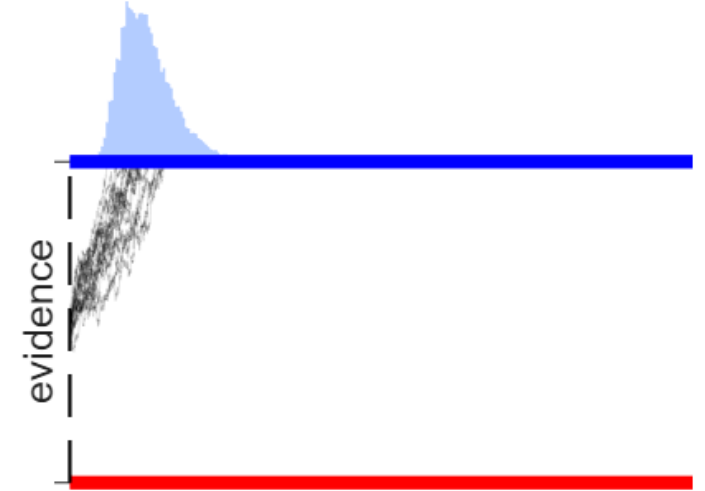
small drift



medium drift



large drift

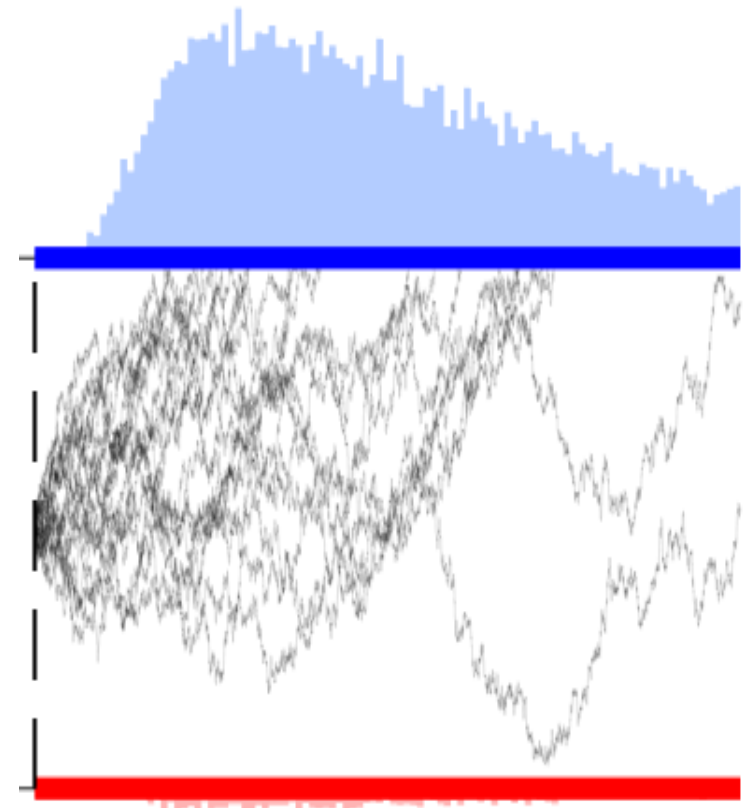
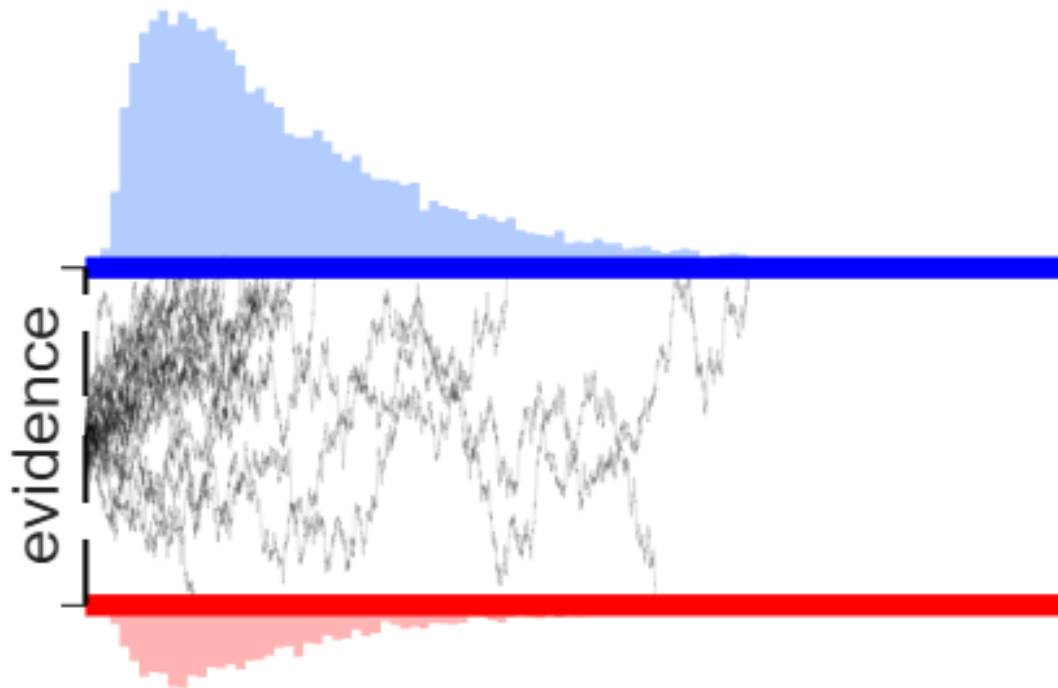


Larger drift leads to more accurate and faster responses.
The drift parameter sets the **sensitivity to evidence**

DDM parameters: the boundary

small boundary

large boundary

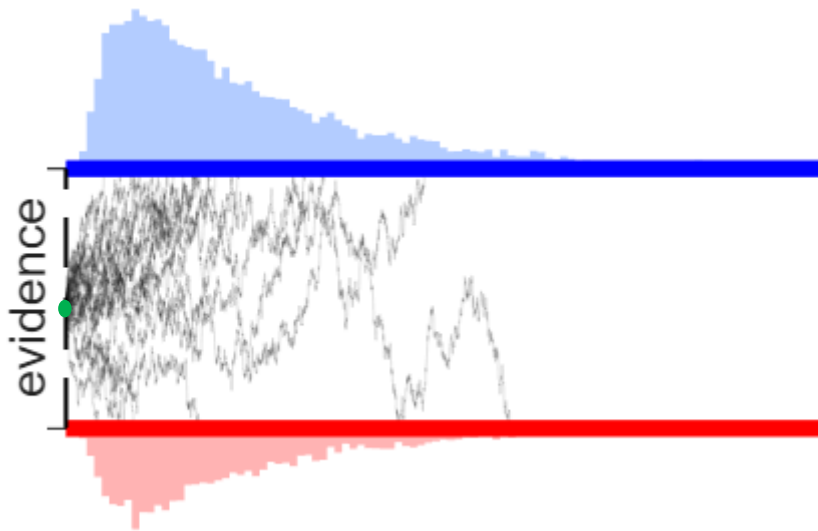


Larger boundaries lead to more accurate but slower responses (and less skew RT distributions).

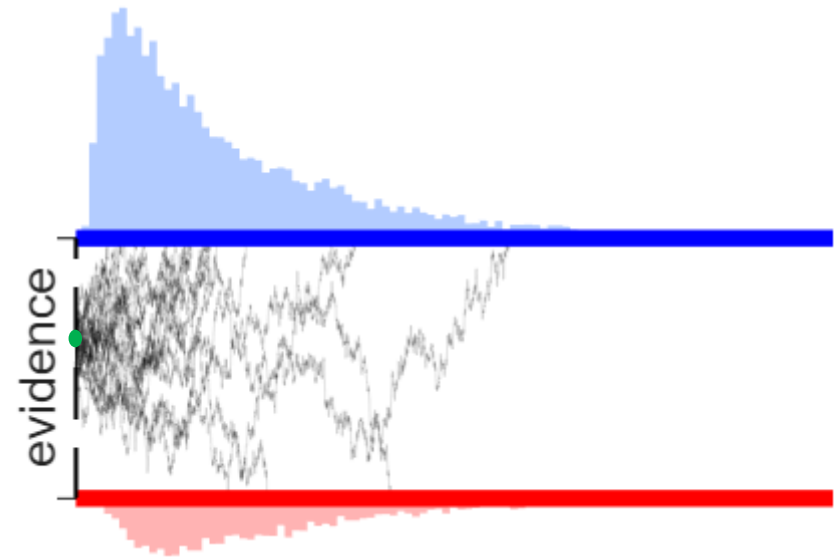
The boundary parameter sets the **speed-accuracy trade-off**

DDM parameters: the starting point

unbiased starting point



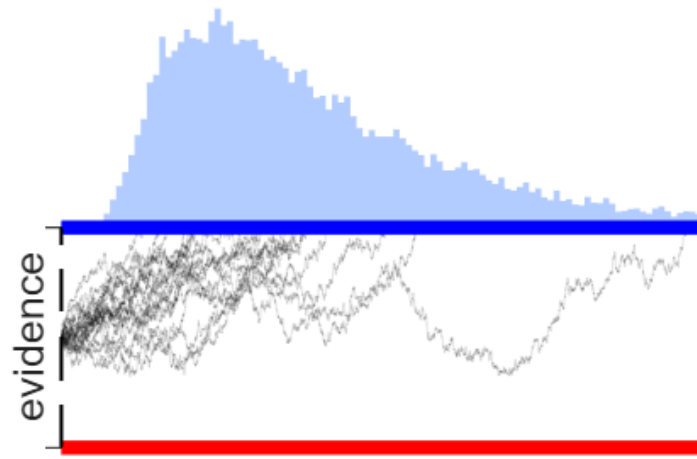
biased starting point



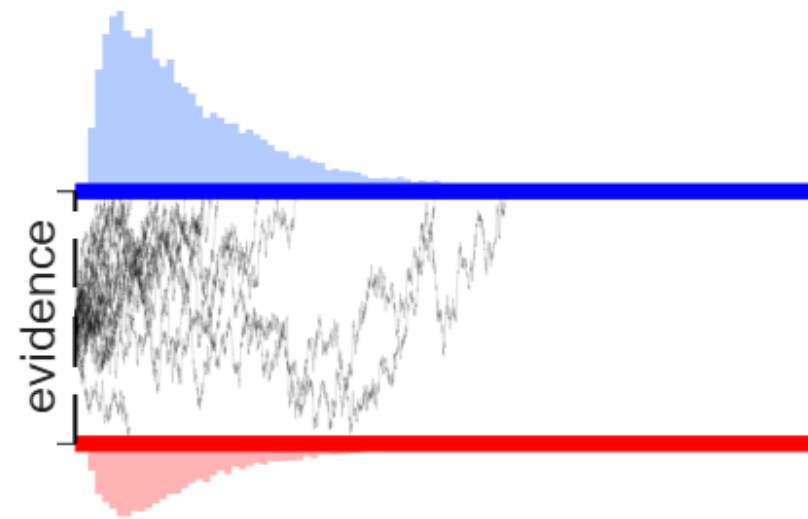
A biased starting point biases choices and speeds up responses on the biased side.
The starting point parameter sets the **prior**

DDM parameters: the noise variance

low noise

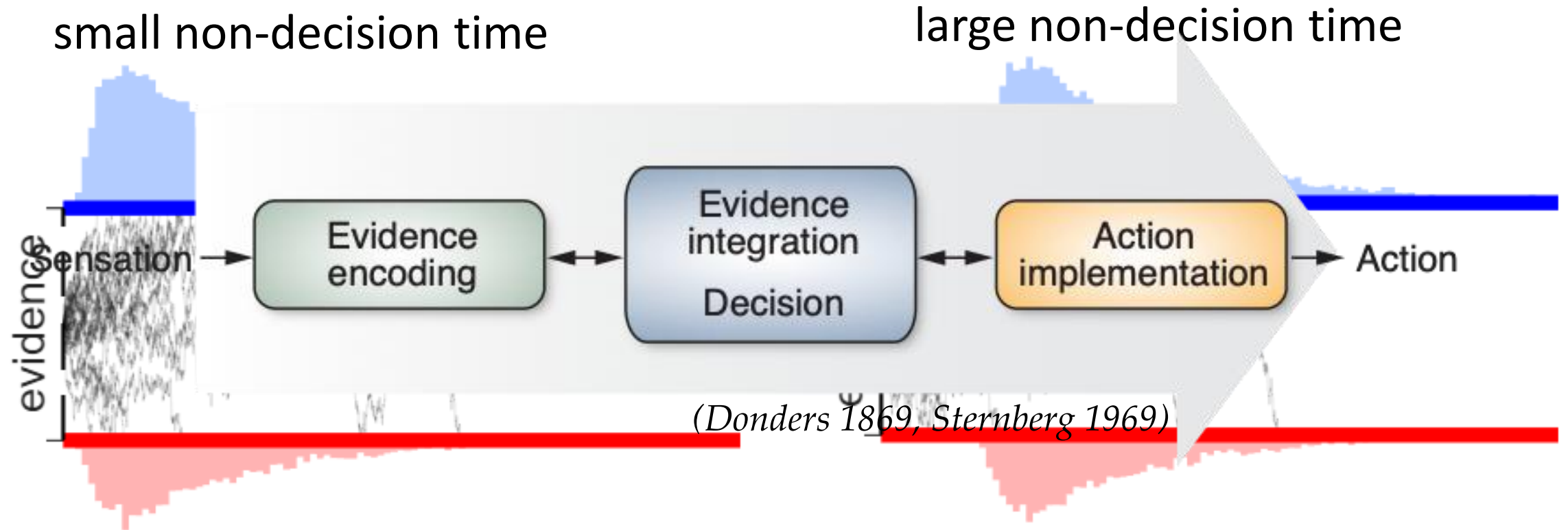


large noise



Larger noise leads to less accurate and faster responses.
It is mathematically **equivalent** to lowering the drift, boundary and starting point
So in practice it is generally set to 1 to make the system identifiable

DDM parameters: the non-decision time

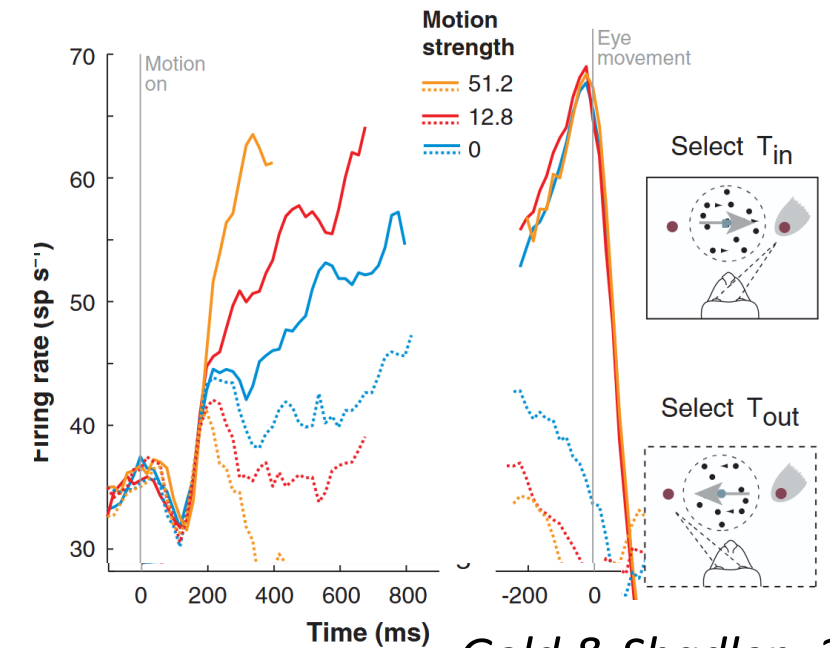


Larger non-decision time simply shifts the RT distribution.
It lumps together **afferent and efferent times**.
It is better estimated by including a condition with large evidence

Empirical support for the DDM

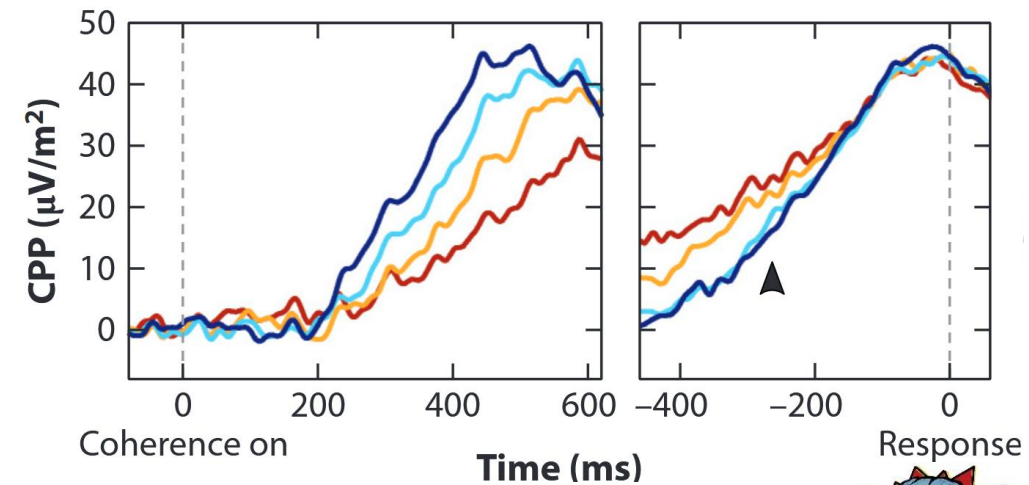
- At the **behavioral** level, the DDM captures choices and RTs in many paradigms beyond simple discrimination tasks (integration over space or modalities, value-based decisions, etc.)
- At the **neural** level, direct evidence of accumulation-to-bound have been found in electrophysiology (LIP) and EEG (CPP)
- The no-decision-before-threshold **policy** could be implemented by cortico-basal ganglia-thalamus loops (in particular STN; *Frank et al. Science 2007*)

Lateral intra-parietal cortex (LIP)



Gold & Shadlen, 2007

Centro-Parietal Positivity (CPP)

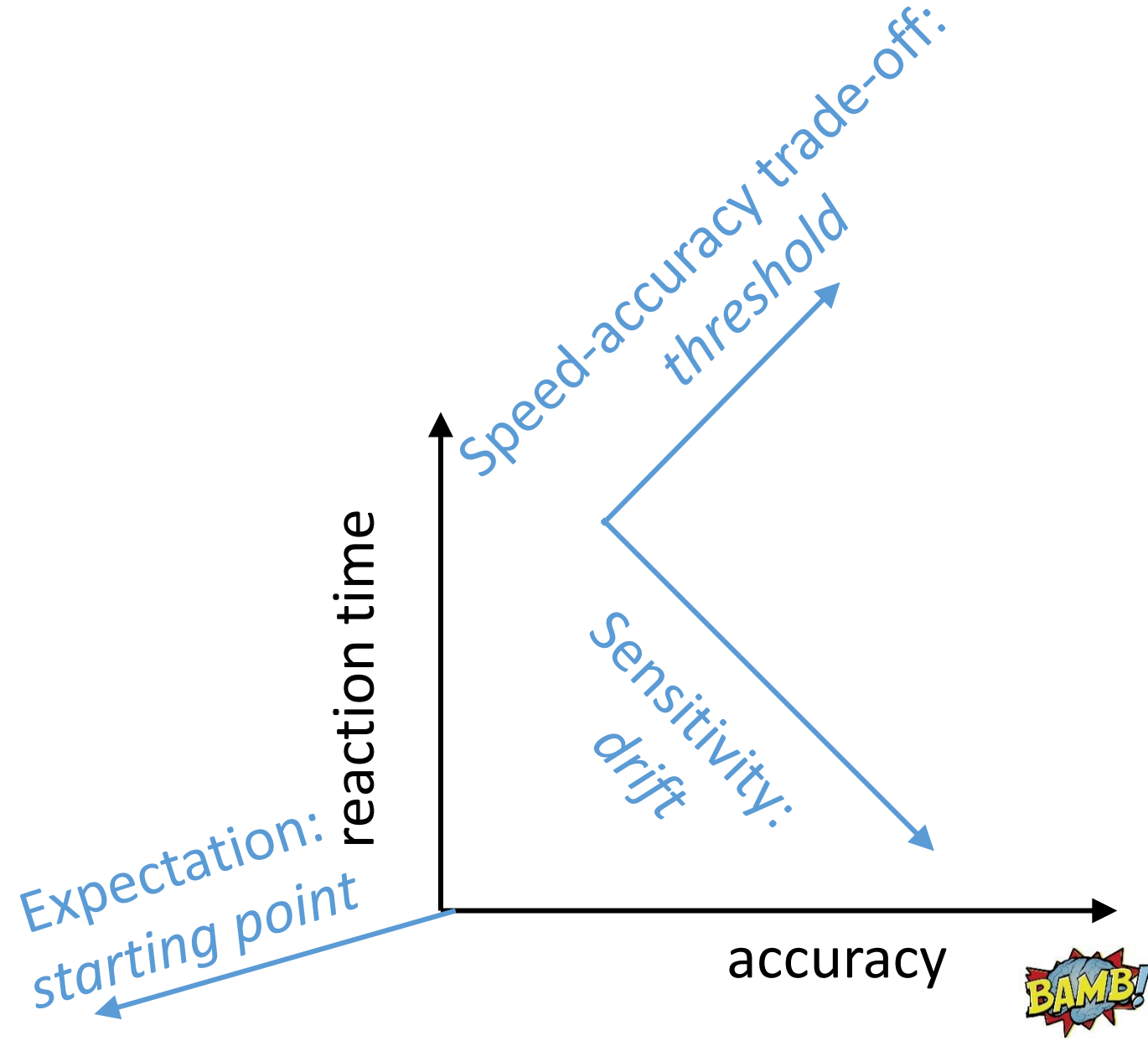


O'Connell & Kelly, 2021



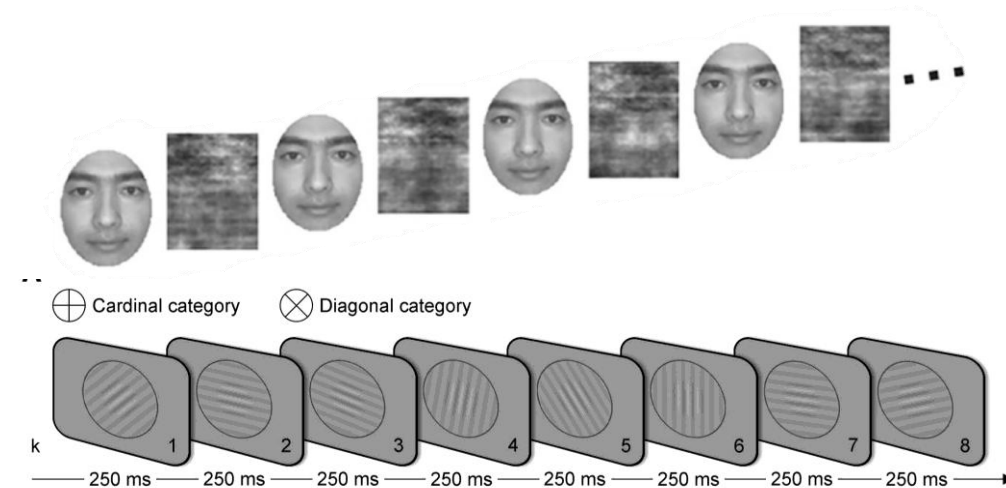
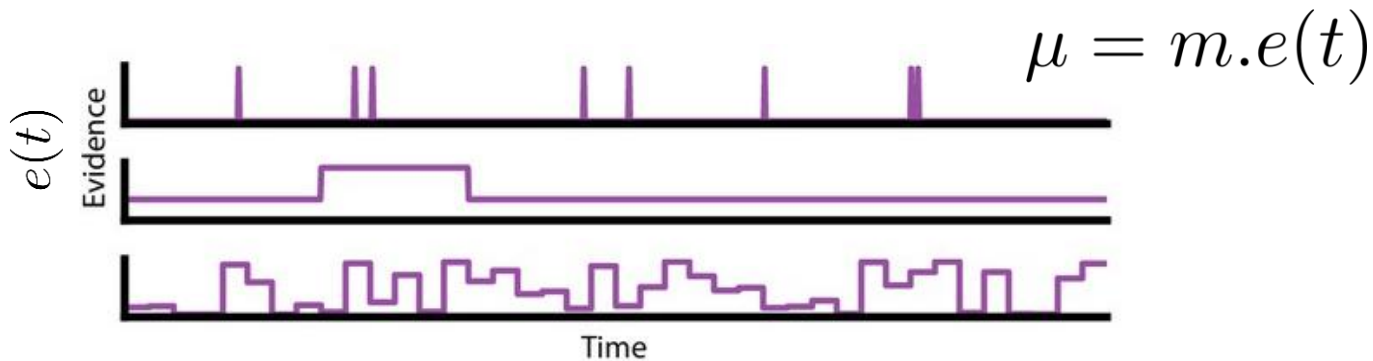
Why you want to use DDM to model your data

- mechanistic approach: Because you think it is an appropriate description for the neural mechanisms of decision-making for your task
- statistical approach: Because you look for a condensed form to analyze the influence of your variable of interest onto choices and RTs jointly (and avoiding possible confounds due to changes in speed accuracy trade-off)

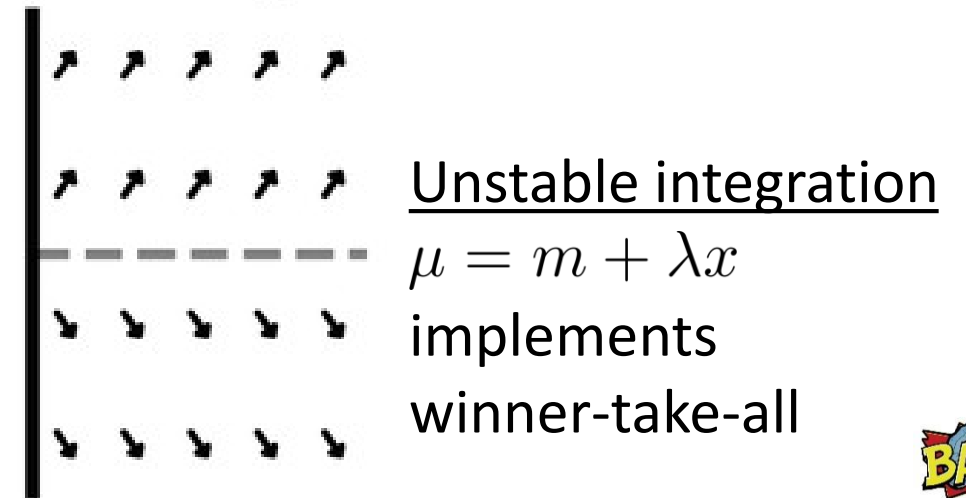
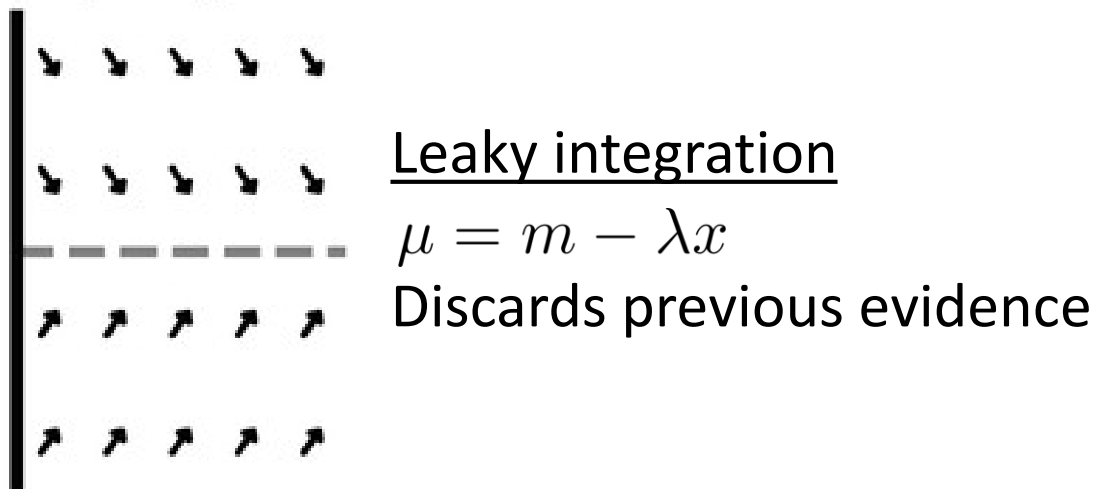


Extensions: changing the drift

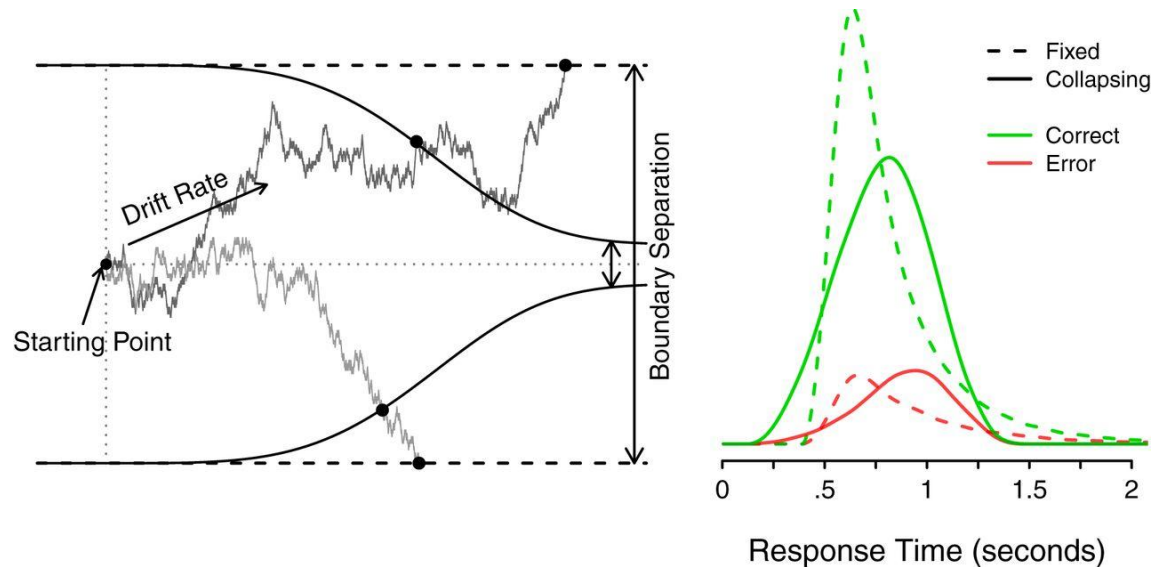
1. Time-varying drift $\mu = \mu(t)$



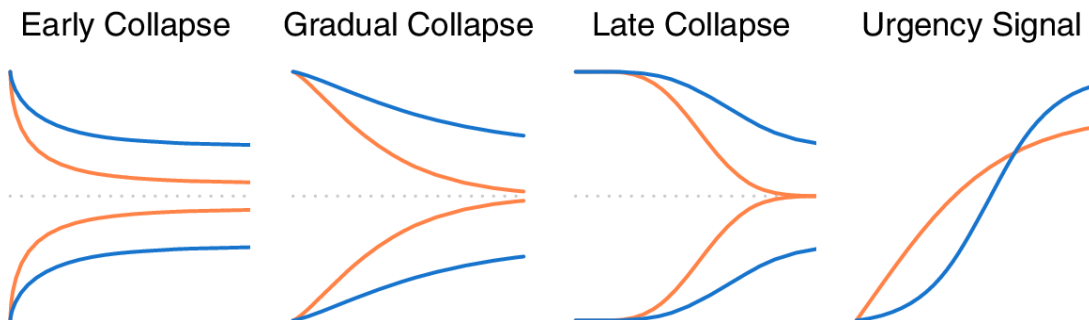
2. State-varying drift $\mu = \mu(x)$



Extensions: collapsing boundaries



- Shorten reaction times, at the cost of accuracy – optimal when evidence varies across trials (*Drugowitsch, Moreno-Bote et al., 2012*)
- Controversial in the literature.. requires additional parameters to be estimated
- inconsistent with the neuroscience (build to threshold signals)
- but equivalent to additive urgency in race models (*Ditterich 2006*), supported by neurophysiological evidence for urgency (*Churchland et al, 2008, Thura & Cisek 2014, Corbett et al, 2023*)



Extensions: incorporating contaminants/outliers

- Very similar to incorporating lapses in psychometric curve
- Small proportion of very fast (premature) or very slow (attentional lapse) responses that were likely not generated by the standard accumulation process.
- MLE is not robust w.r.t to such outliers, so we need to incorporate those outliers with some fixed proportion λ (from 0.1% to 5%) to avoid biasing the estimates.

$$p(c, RT; \boldsymbol{\theta}_{DDM}, \lambda) = (1 - \lambda)p(c, RT; \boldsymbol{\theta}_{DDM}) + \lambda p_{outlier}(c, RT)$$

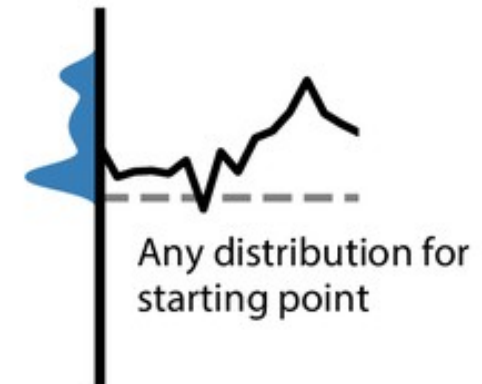
(Ratcliff and Tuerlinckx, 2002)



Extensions: trial-to-trial variability

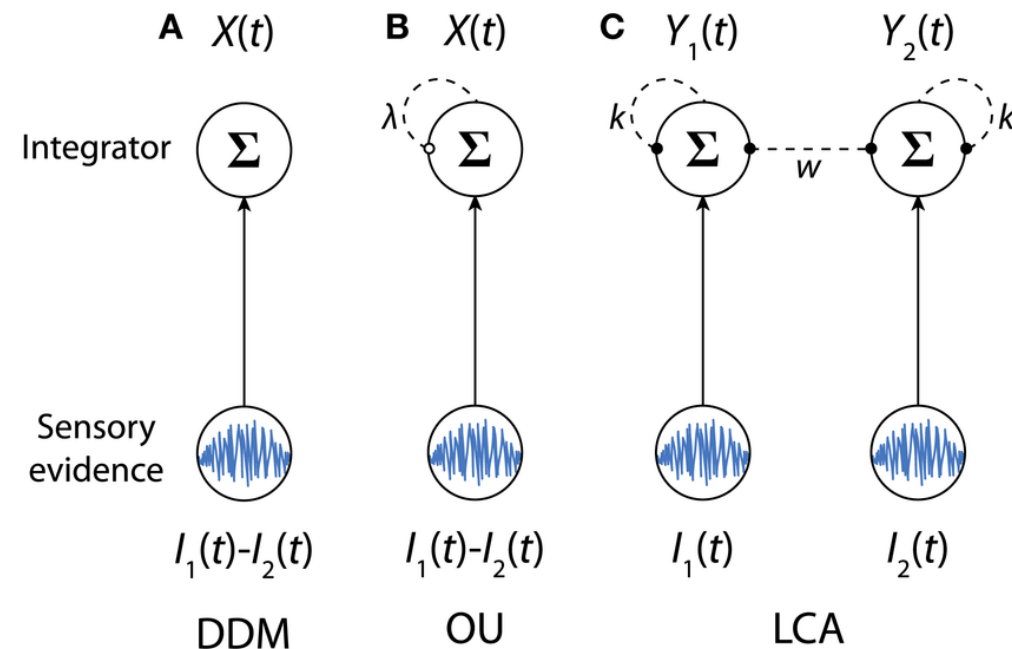
- DDM parameters (drift, threshold, starting points, non-decision time) may change over trials
- Trial-to-trial drift variability can account for slower errors
- Trial-to-trial threshold variability can account for premature errors
- Can make fitting more complex or even non-identifiable
- This variability can be **linked to brain function** by using neural signal as regressor to DDM parameter

Any distribution for non-decision time



Limitations of the DDM

- Free-response: RT may be non-informative
- Even if free-response, the timing of the response may not depend on accumulated evidence (*Hawkins & Heathcote, 2021; Hernández-Navarro et al., 2021*)
- Limited to two-alternative forced choices:
 - If just one, same framework but with single bound (Wald distribution)
 - If more than two, one accumulator per possible response (*race models, Leaky Competing Accumulators, Linear Ballistic Accumulators*)



Zhang et al., 2012



Sequential sampling models

Relative evidence criteria: Random walk/diffusion models

Continuous time
continuous
evidence:
**Diffusion
models**

Decay in drift
**Ornstein Uhlenbeck
(OU) model**

Constant drift
**(Standard) Wiener
Diffusion model**
aka 'Drift Diffusion Model

Discrete time
continuous
evidence:
**Random
walk models**

Absolute evidence criteria: Accumulator models

Discrete time
discrete
evidence:
**Recruitment
model**

Discrete time
continuous
evidence:
**Accumulator
model**

Continuous
time, discrete
evidence:
**Poisson
counter Model**

Continuous time,
continuous evidence:
non-stochastic
**Linear ballistic
accumulator model**

Hybrid accumulator/diffusion models
absolute evidence criteria.

Independent evidence totals
with or without decay in drift
Dual diffusion model

Inhibition between evidence totals,
decay in drift: **Leaky competing
accumulator (LCA) model**

How to fit sequential sample models

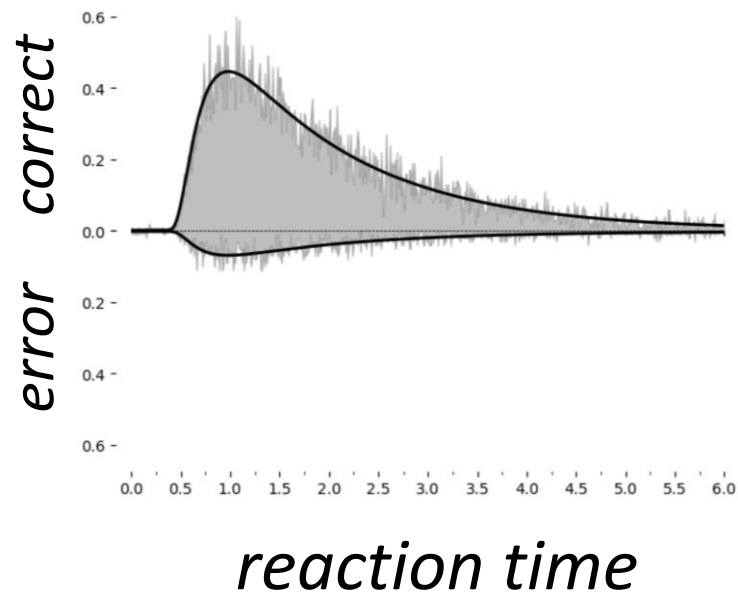
- **Vanilla DDM:** the likelihood can be computed analytically
- **Extended DDM:** the likelihood can be computed using the Fokker-Planck equation (*Shinn et al. eLife, 2020*)
- **Other models:** in general no analytical formula for the likelihood
 - classical method: error on RT quantiles
 - modern method: approximating the likelihood using *simulation-based inference* (i.e. in MNLE, *Boelts et al. eLife, 2022*)



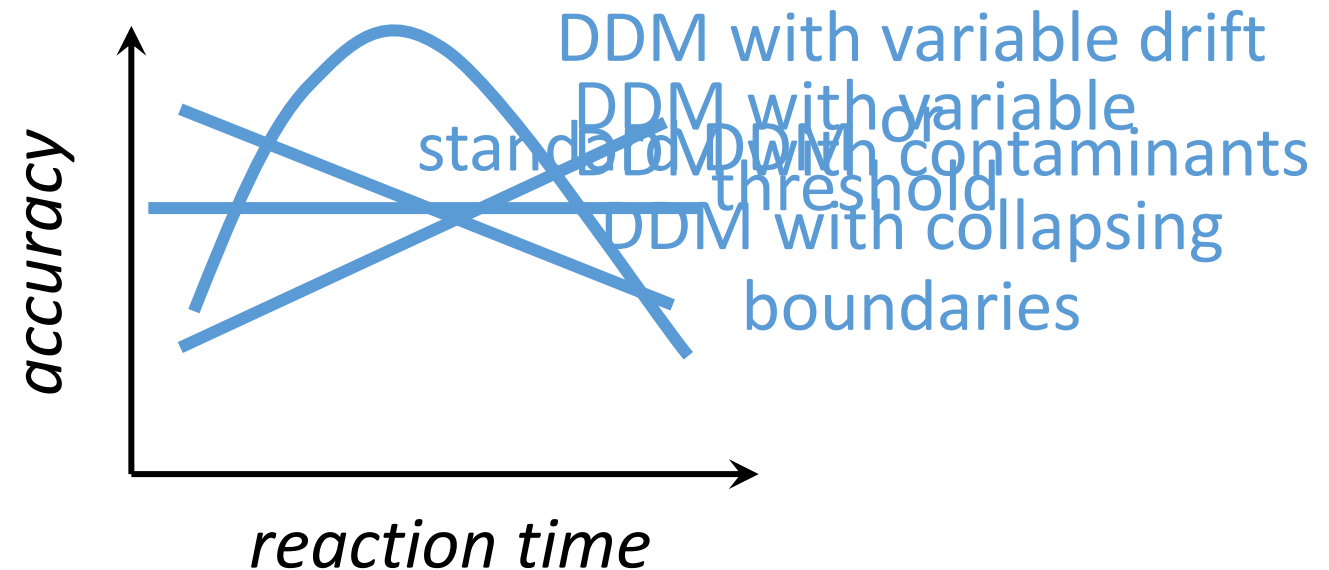
Validating the DDM on behavioral data

It is **crucial** to check that the fitted model captures the important features in the behaviour!

Reaction Time densities

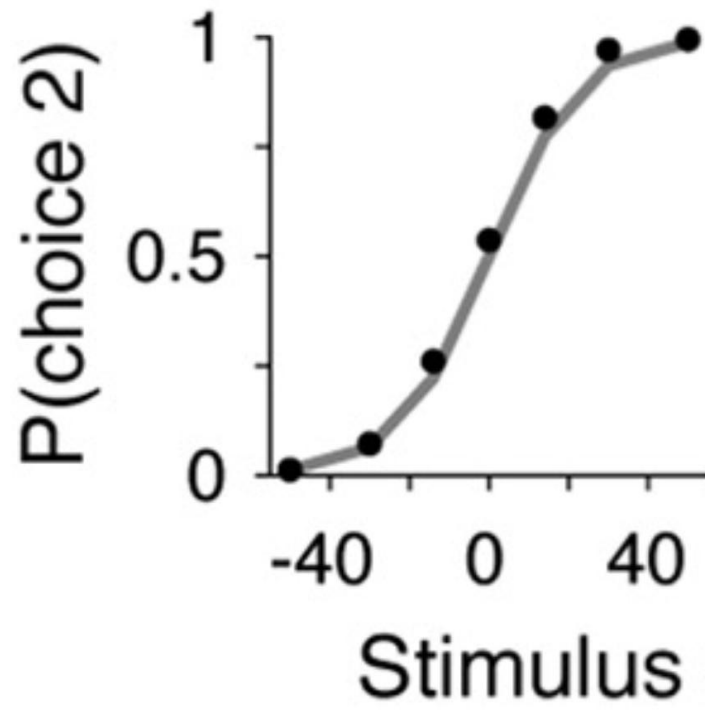


Tachometric curve

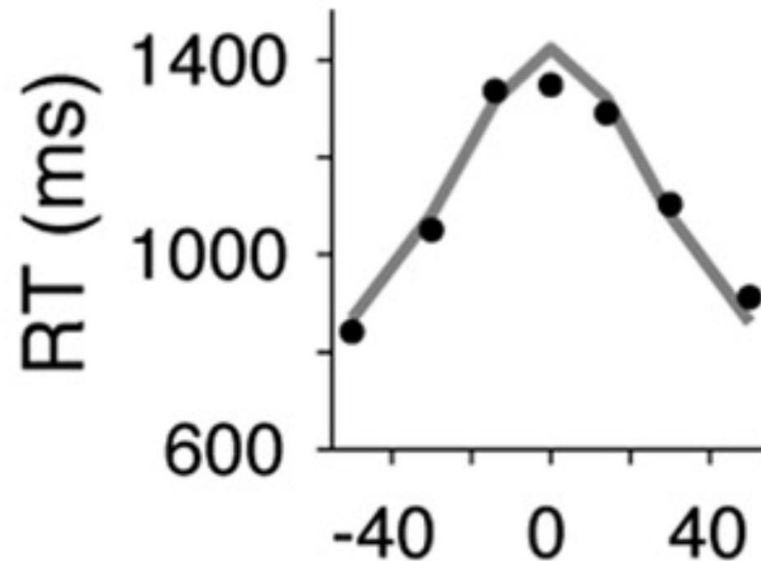


Validating the DDM on behavioral data (varying evidence)

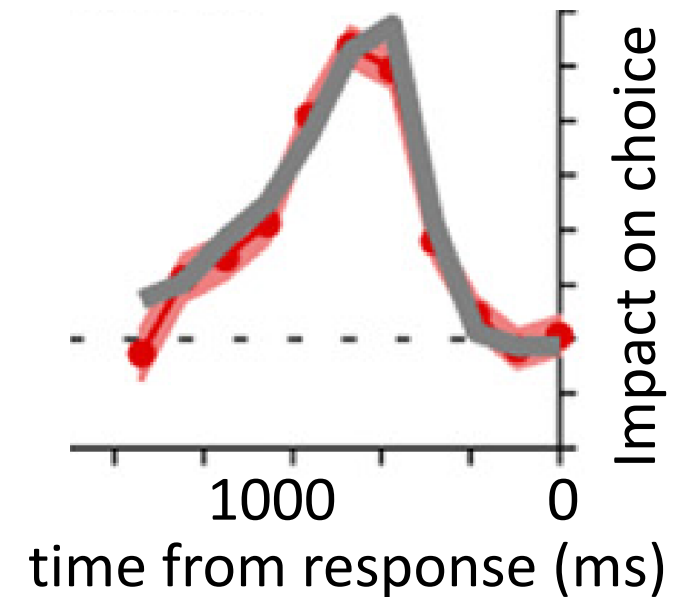
Psychometric curve



Chronometric curve



**Psychophysical kernel
(time-varying evidence)**



Summary

- The Drift-Diffusion Model is very popular for fitting choices and reaction times jointly as it is...:
 - **theoretically** well-grounded (link to SPRT)
 - **empirically** supported both at the behavioral level (captures choices and RTs in a wealth of settings) and neural level: LIP, CPP
 - **practically** easily fitted to data: analytical formula for likelihood, implemented in various packages (*PyDDM*, *hssm*,...)
- It can be **extended** to better capture behavior (trial-to-trial variability, collapsing bounds, leaky integration, etc.)
- As any model it has its own **limitations** (two choices only, free reaction time) so using it must be well justified a priori and validated a posteriori





The Drift- Diffusion Model – the tutorial

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Module 1

Part 1: Simulating the DDM by hand

- Basic algorithm

- 1. Set x to starting point

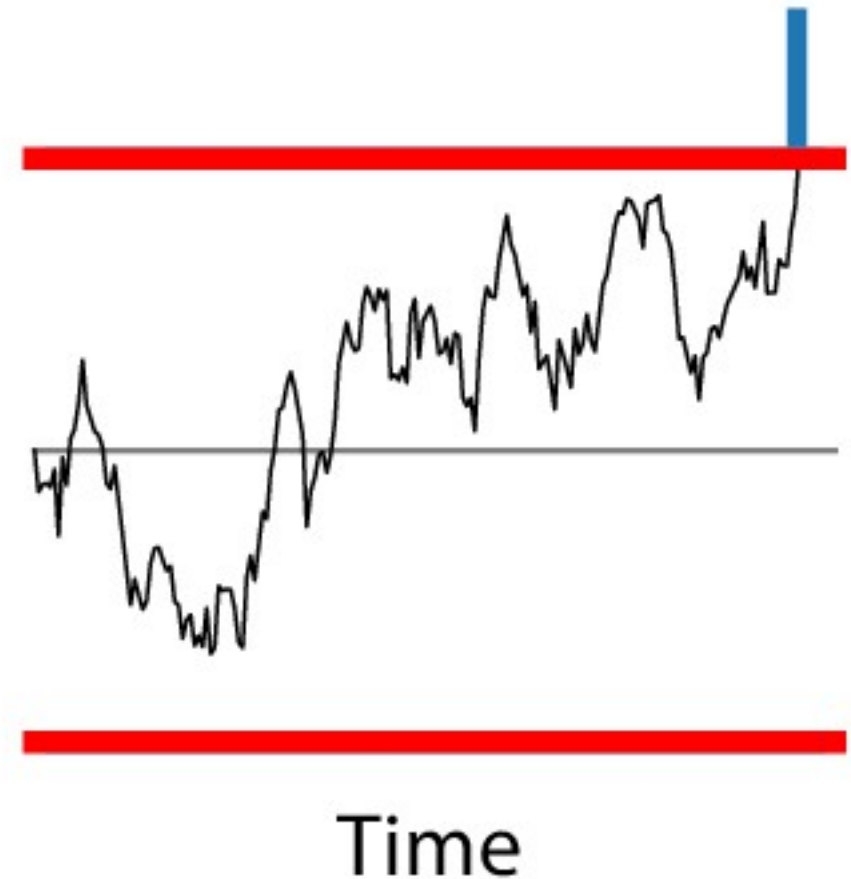
- 2. Set:

$$x_{t+1} = x_t + [\text{drift}]\Delta t + [\text{noise}]z_t\sqrt{\Delta t}$$

$$z_i \sim N(0, 1)$$

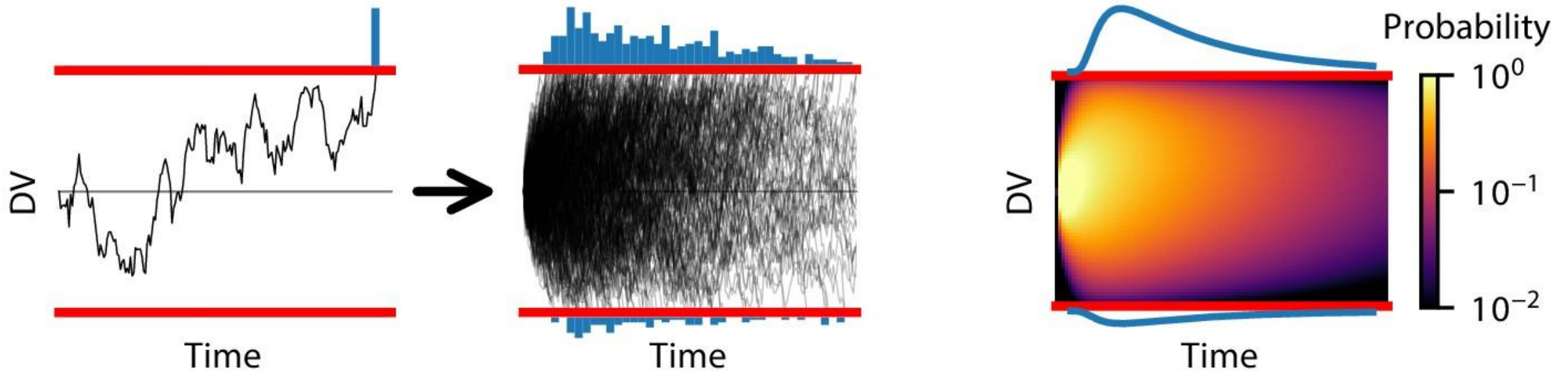
- 3. Check if x crosses a boundary. If so, you are done

- 4. Otherwise, go to (2)



Part 2: Simulating the DDM using PyDDM

- Use more efficient methods to simulate the probability distribution of a trajectory's position instead of one trial at a time



How PyDDM works:

- Construct a Model from its components
- Model components:
 - Drift rate
 - Noise
 - Bound
 - Initial Condition
 - Non-decision time
 - Mixture model



Many model components are built-in:

- Each component can be:
 - A constant value (e.g., 3)
 - A fittable parameter, given by the the name (e.g., `param1`)
 - A function which depends on:
 - Parameters
 - Task conditions
 - Magic arguments



Parameters and conditions

- Parameters: Have the same value for the entire dataset
 - E.g. bound height
- Conditions: May change from trial to trial
 - E.g. strength of motion coherence



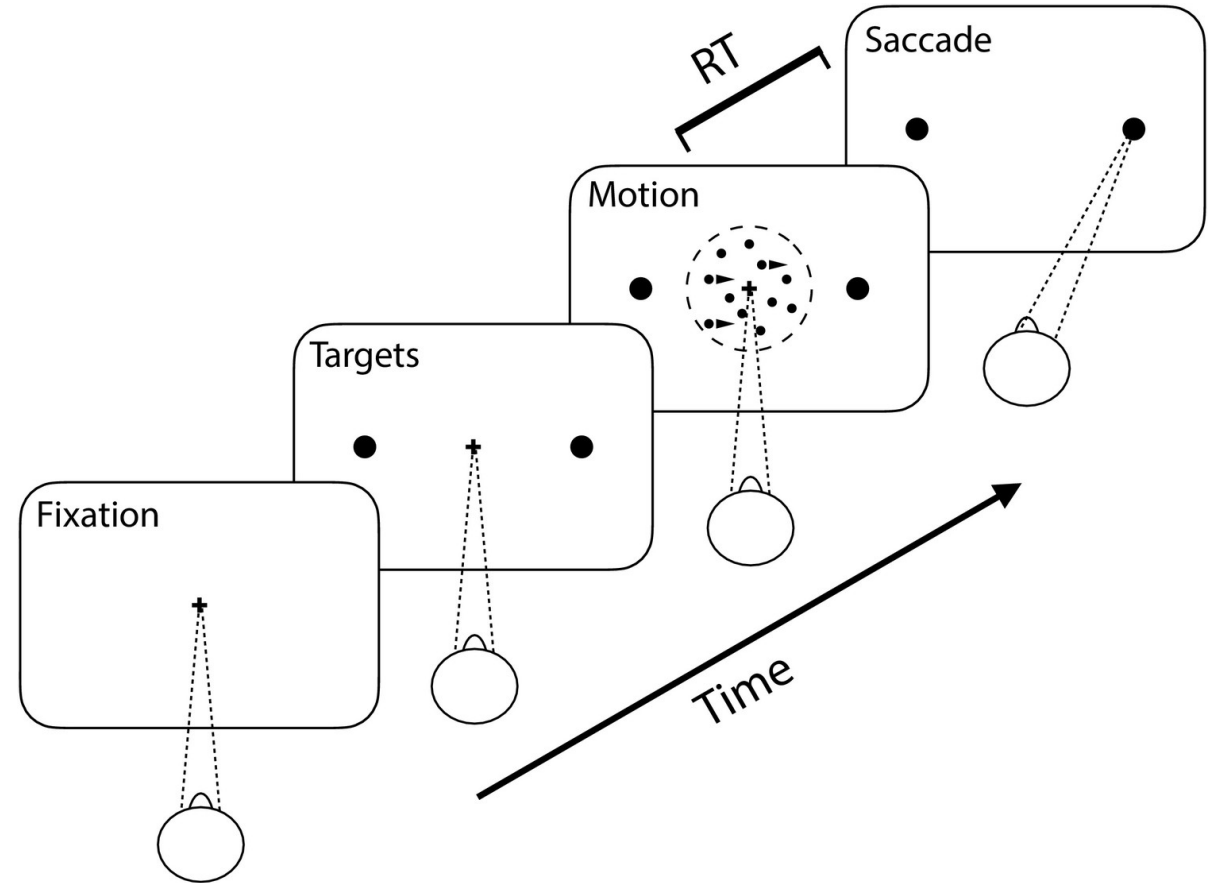
Three objects to remember in PyDDM

- Model: created by the `gddm()` function
 - May need to call `model.fit()` before using if there are parameters
- Solution: Created using `model.solve(conditions={...})`
- Sample: RT and choice data, either experimental or simulated data



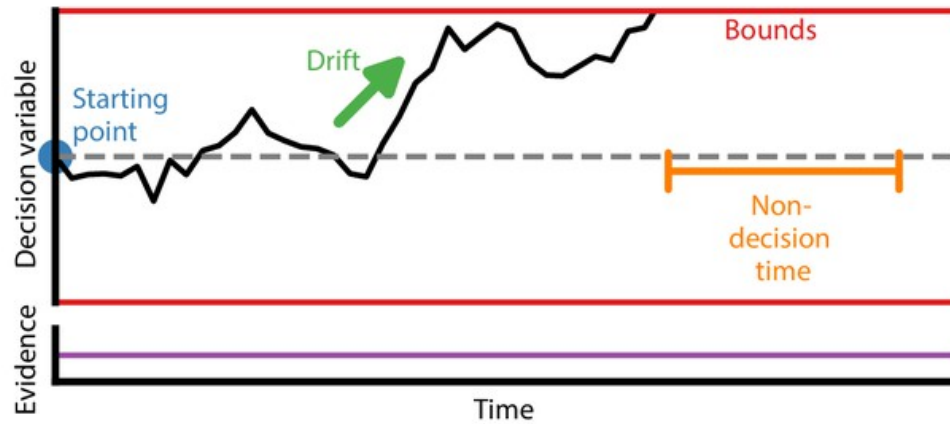
Part 3: Fitting the DDM to data

- Dataset: Monkeys performing the random dot motion task (Roitman and Shadlen, 2002)
- Several levels of motion coherence

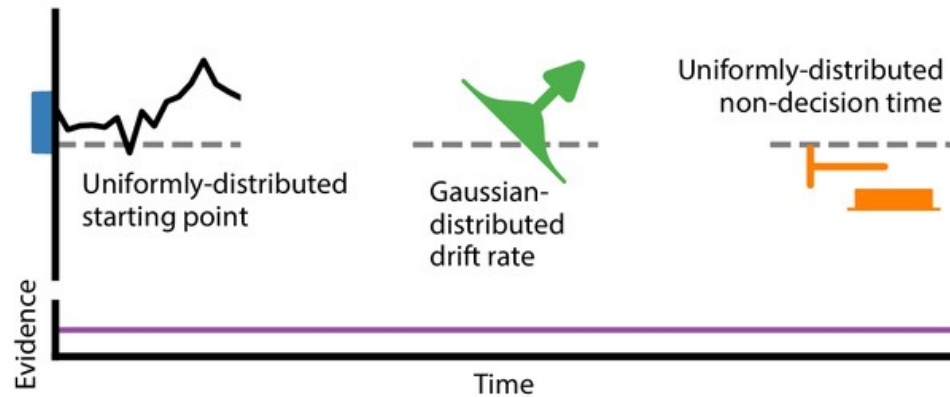


Generalized DDM (GDDM)

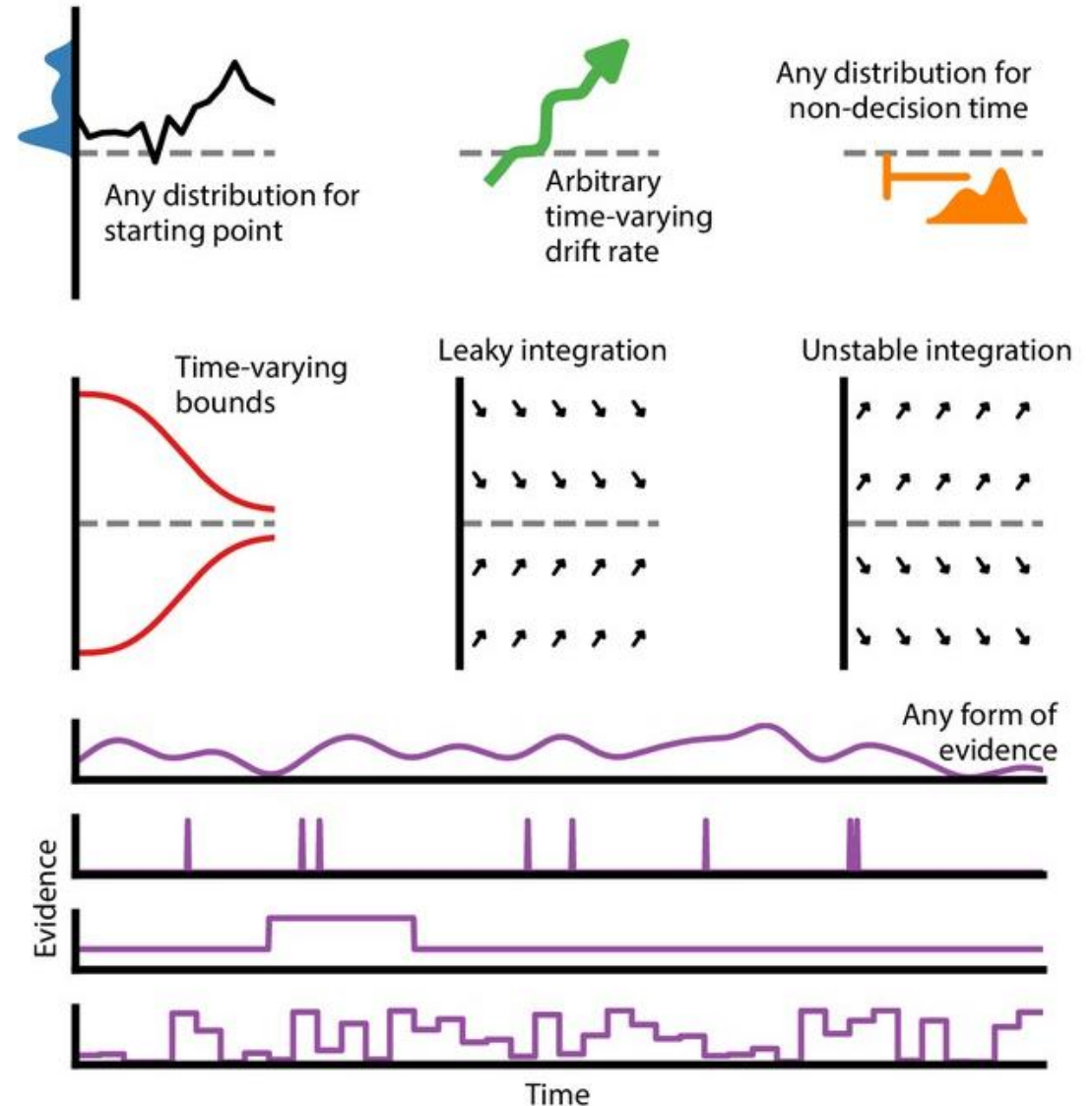
DDM



Full DDM



GDDM (examples)



When should you use these GDDMs?

- Response time distribution is not skewed
 - Consider leaky integration
- The speed-accuracy tradeoff may change across the trial
 - Consider collapsing bounds
- I think the agent may be more likely to choose one choice over another or have a prior
 - Consider a starting point or drift bias
- Evidence is not constant in my task or it requires multisensory integration
 - Consider a more complex drift rate function
- There is a large variability in motor actions
 - Consider non-decision time variability (*but be careful! This can make the model non-recoverable*)



Accuracy coding vs Stimulus coding

- **Accuracy coding:**

Upper boundary: correct choice

Lower boundary: incorrect choice

Drift rate should be positive (if subjects are above chance)

Very practical to use

- **Stimulus coding:**

Upper boundary: one possible response (say, rightwards)

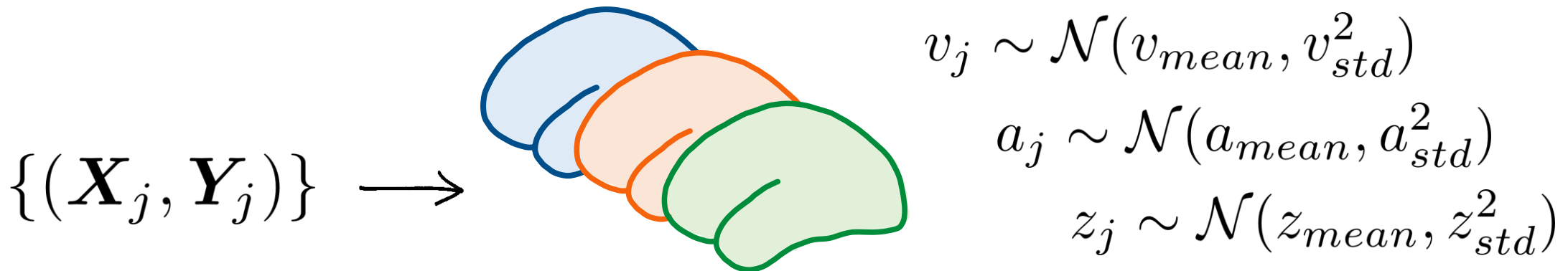
Lower boundary: the other possible response (leftwards, then)

Drift rate should be signed with the identity of the stimulus (positive if stimulus associated with right, negative if stim associated with left)

Mandatory if you want to include bias towards one side (starting point no midway between the two boundaries)



Hierarchical models (or mixed models)



HDDM directly estimates mean and std for each parameter across the population (also provides estimates for single subject)

If you want one parameter to be fixed (no variability across subjects):

```
model = hddm.HDDM(data, include=['sv', 'st'], group_only_nodes=['sv', 'st'])
```

HDDM uses prior for these parameters based on what's been found in the literature (MatzkeWagenmakers09)